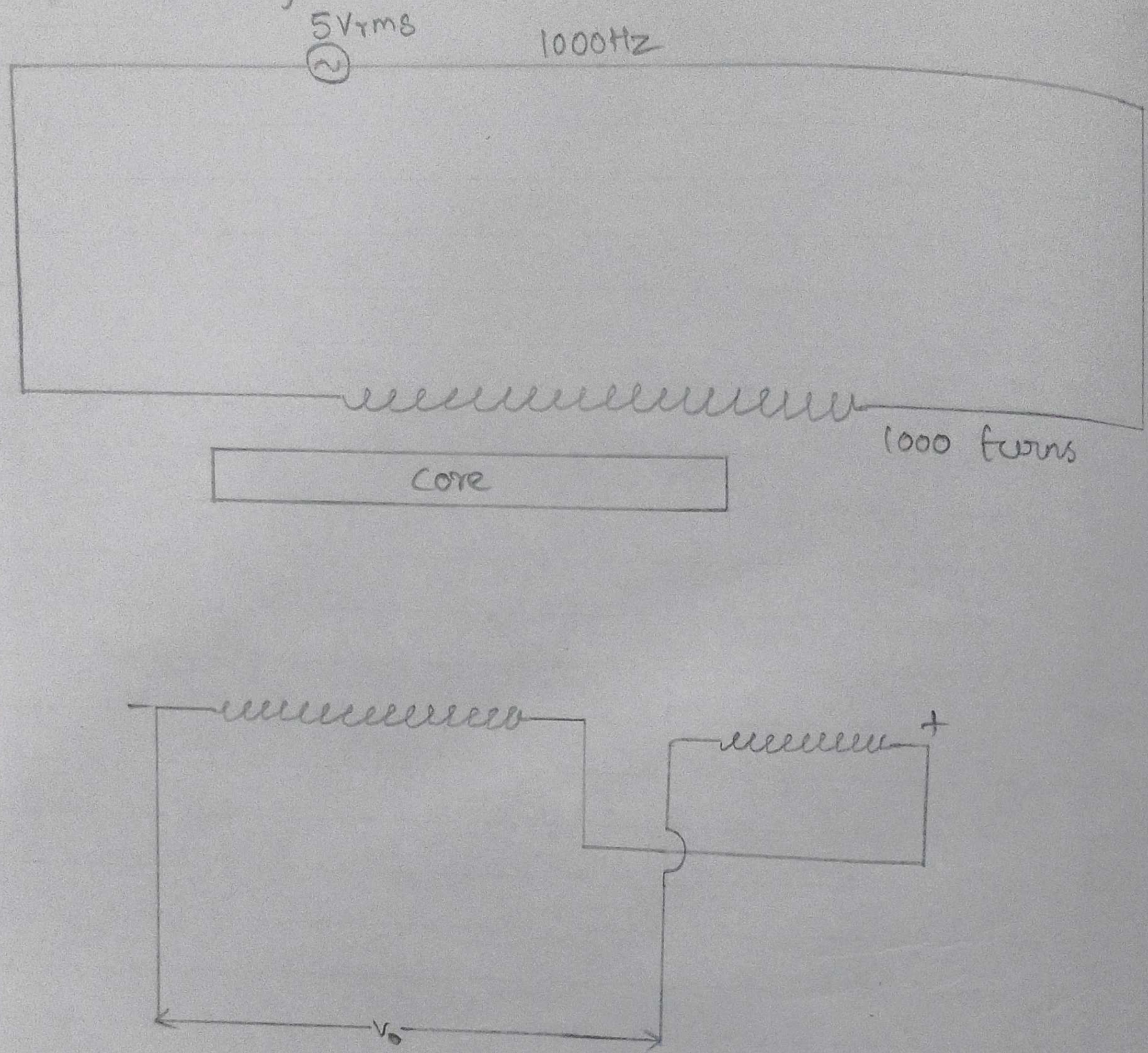
Circuit Diagram:




Tabulation:

| Input voltage (V) ($V_m$) | | Output Votage | frequency (Hz) | Time = 1/8. |
|---|---|---|---|---|
| Positive Displacement | 7 | 2.2V<br>D = 5mm | 1000 | $\frac{1}{1000}$ = 1ms |
| Negative Displacement | 7 | (D = -5mm) | 1000 | 1ms |

# Characteristics of LVDT
## (Linear variables differential Transformer)

**Aim:**

To understand and simulate the relation between core displacement and output voltage.

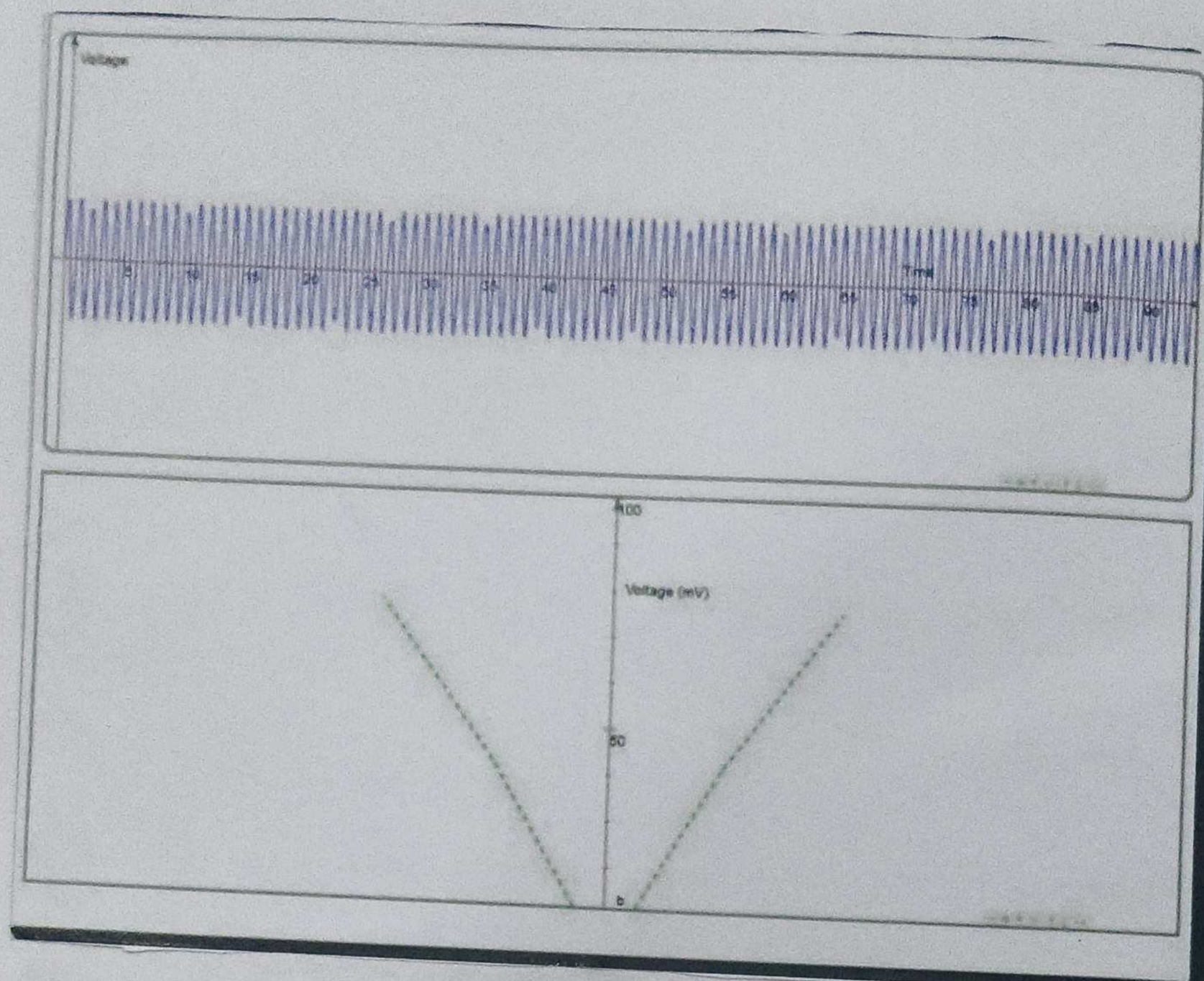
**Apparatus Required:**

Laptop with internet connection.

**THEORY:**

LVDT is an inductive transducer that conneverts linear displacement into a electrical Siginal. It consist of a transformer having one primary windings and having Secondary windings. wounded on a core. As the primary is connected to an ac Source, the ac current and the voltages are produced in the Secondary of LVDT

When the core with both the center, the flux linking with both the Secondary winding are equal. So the emf induced in both So the windings are equal, this means there is no displacement. When the core is at the right, the flux linking with the secondary windings $S_1$ is more So, the emf induced in $S_1$ is more than $S_2$. when the core is at the left, the


Voltage
Time
100
Voltage (mV)
50
0

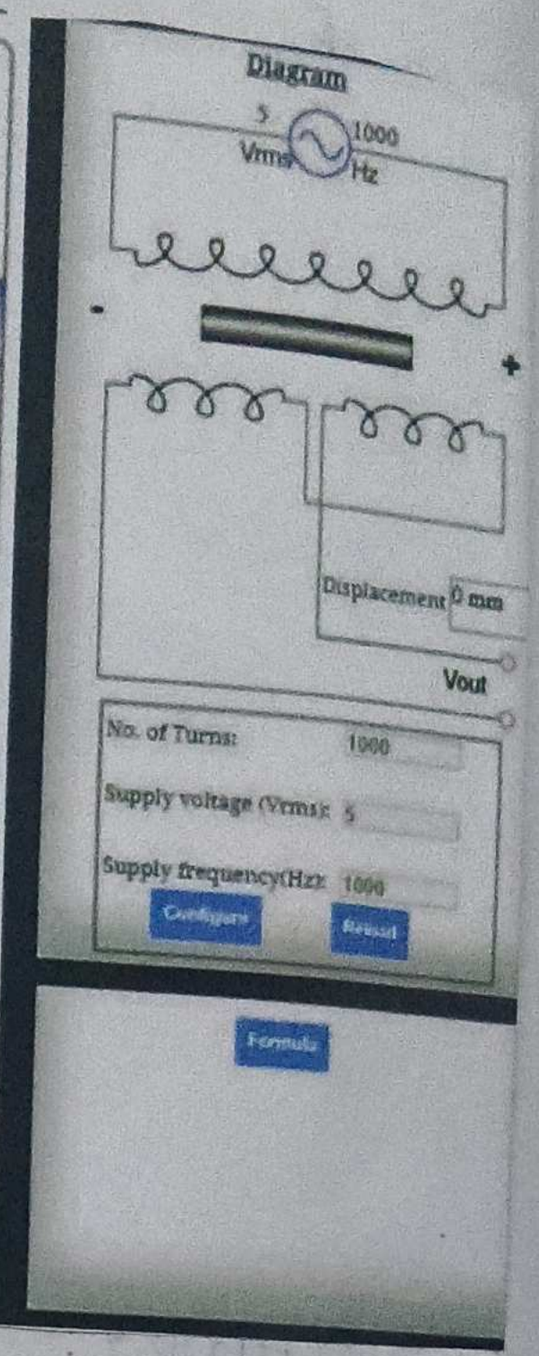
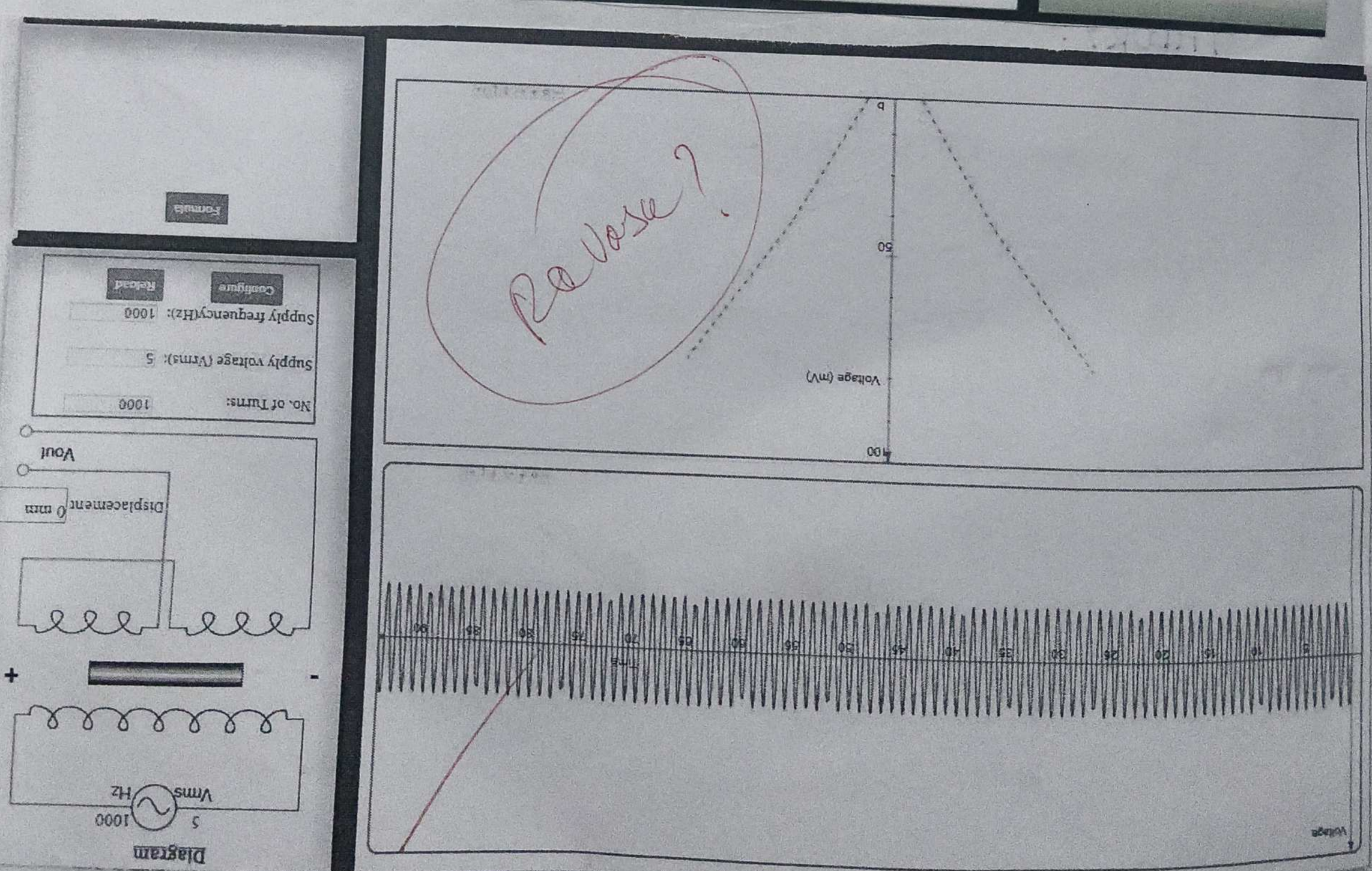


Diagram
5
Vrms
1000
Hz
-
+
Displacement
0 mm
Vout
No. of Turns:
1000
Supply voltage (Vrms):
5
Supply frequency(Hz):
1000
Configure
Reload
Formula


Voltage
Time
100
Voltage (mV)
50
0
Re Verse?
Diagram
5
Vrms
1000
Hz
-
+
Displacement
0 mm
Vout
No. of Turns:
1000
Supply voltage (Vrms):
5
Supply frequency(Hz):
1000
Configure
Reload
Formula

the flux linking and the emf in $S_2$ is more than $S_1$
So, the net emf is negative

PROCEDURE:

* Connections are made as per the circuit diagram.

* Set the number of turns supplied voltage, and Supplied frequency.

* Configure the parameters.

* Move the core to positive side and the negative side, and observe the input voltage and output voltage waveforms.

* plot the graph between input and output voltage

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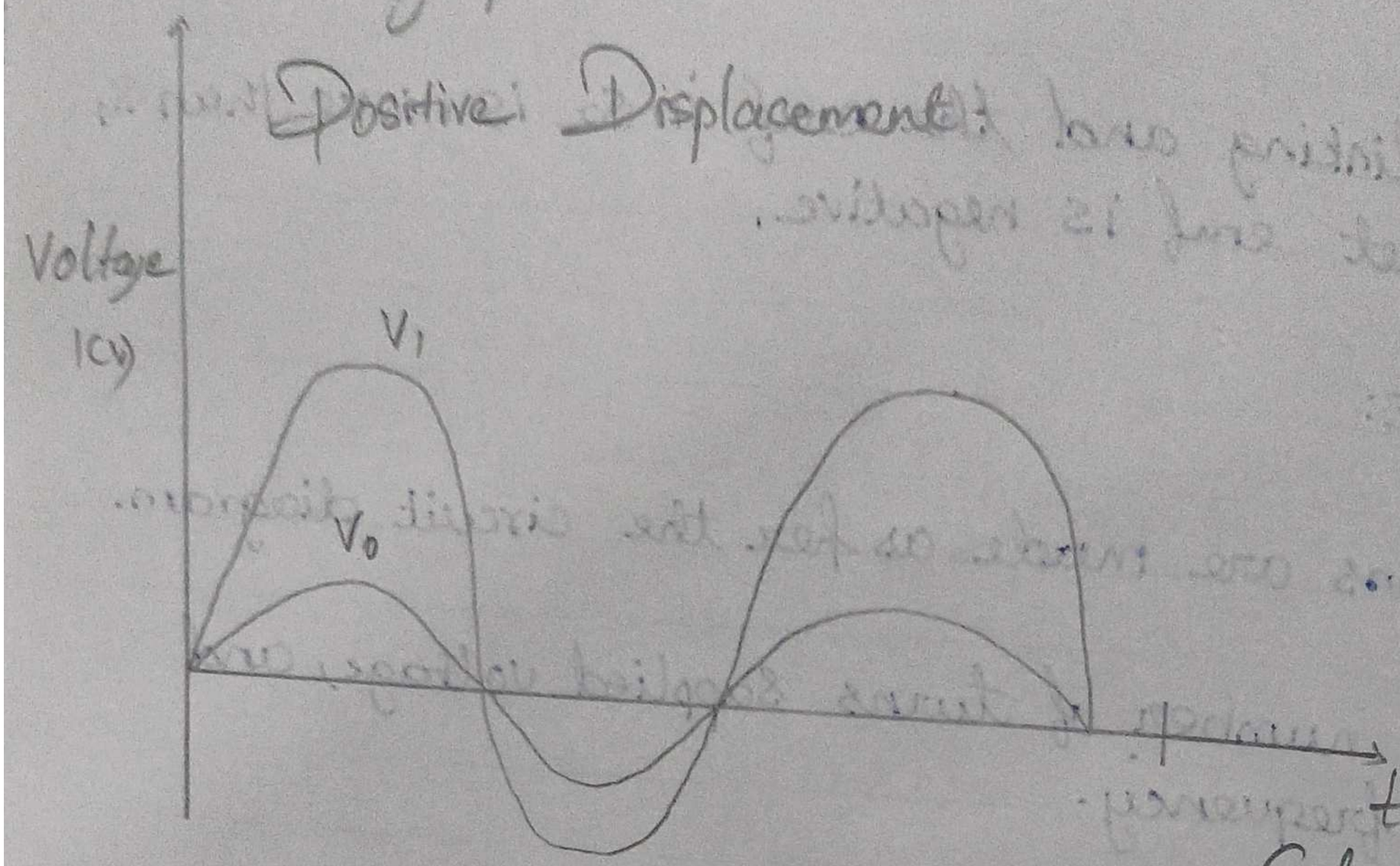
Model Graph:

Positive Displacement:




Calculation

$V_m = \sqrt{2} \times V_{rms}$

$\sqrt{2} \times 5$

$= 7V$

$T = 1ms$.

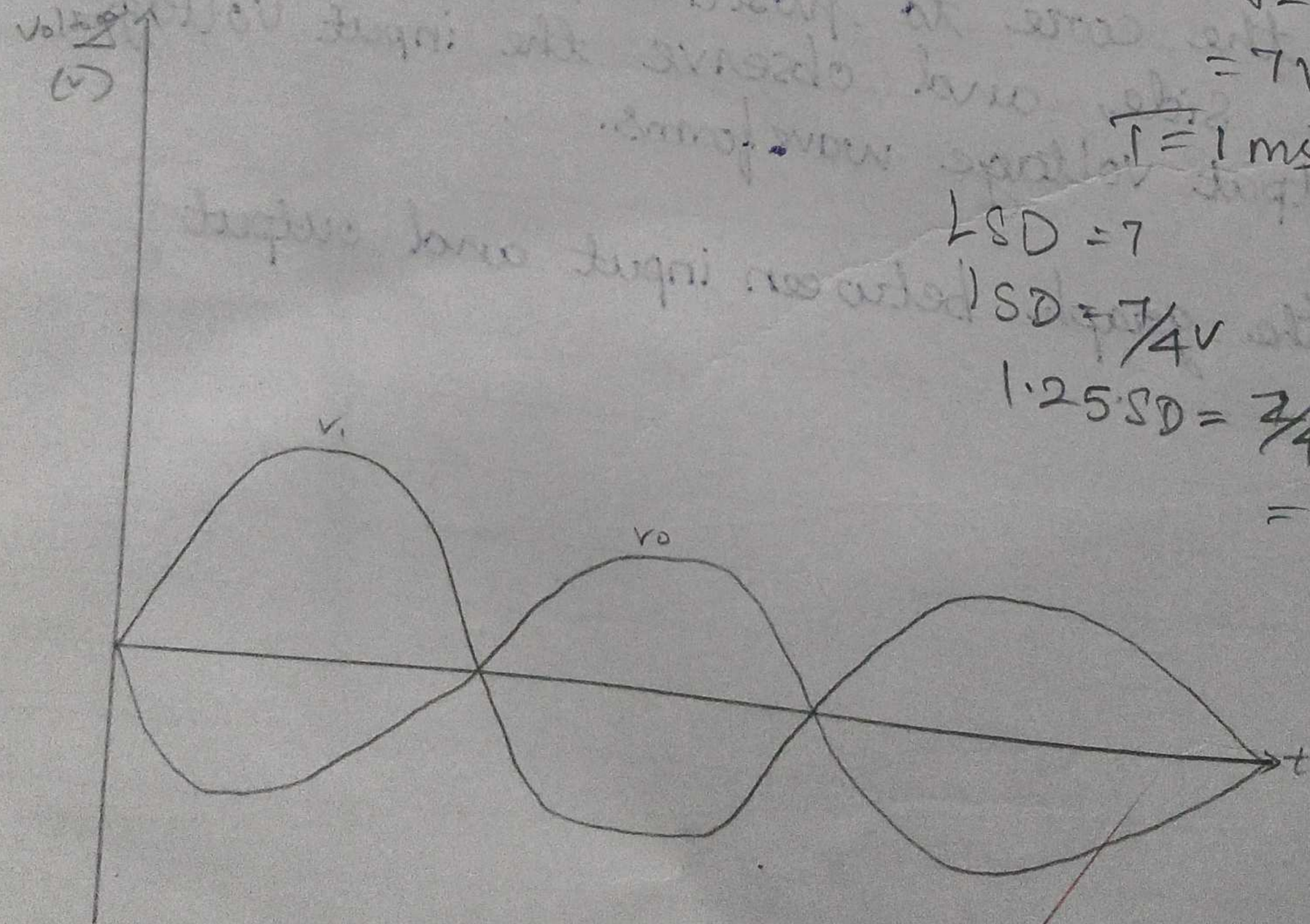
LSD = 7

1SD = 7/4 V

1.25 SD = 7/4 × 12.5

= 2.2 V

Negative Displacement



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Page No.

98

18/03/24

**Result:**

To understand and Simulate the relation between core displacement and output voltage were developed and tested successfully.